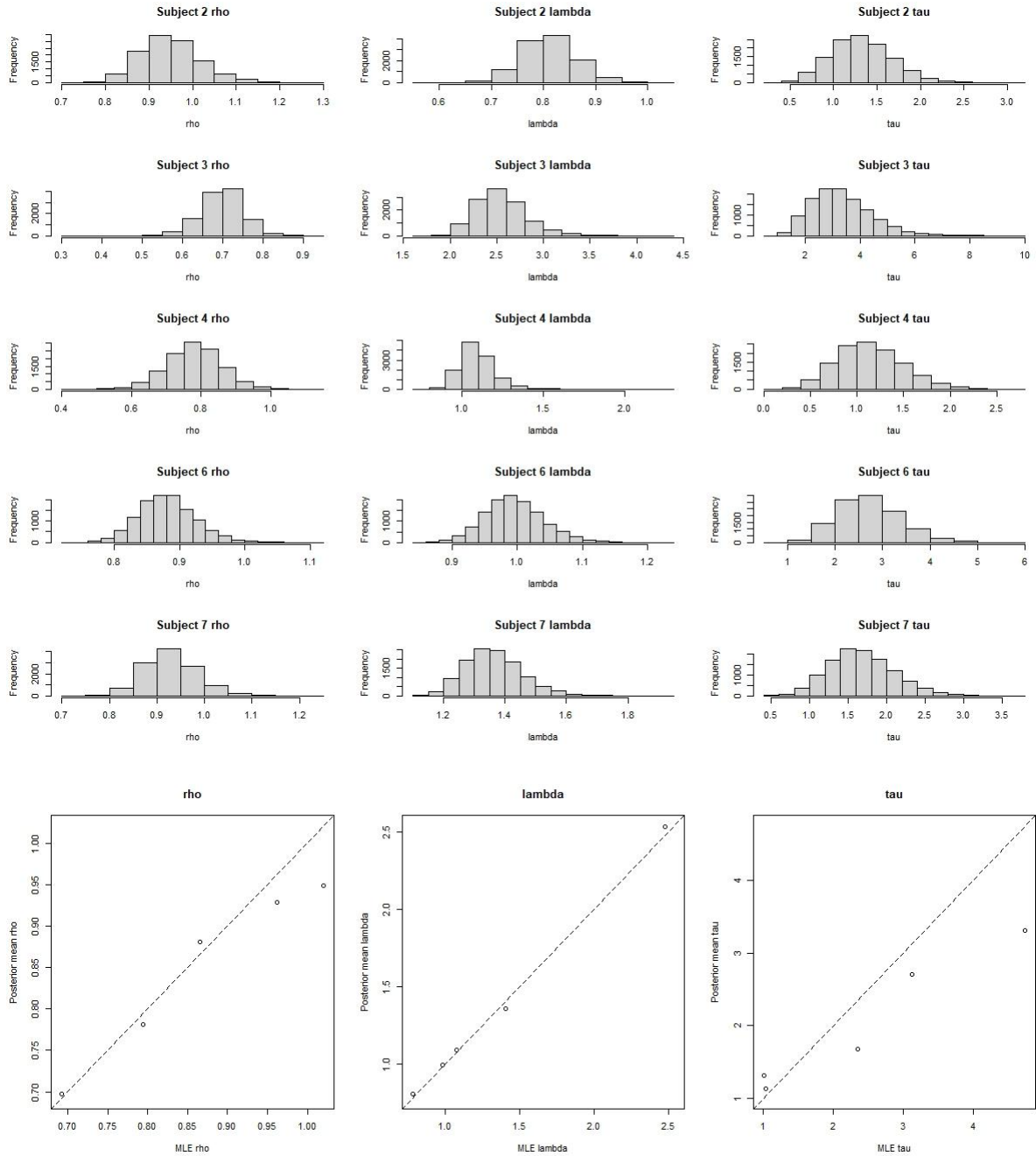


HW5 report

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1. (1)



1. (2) Overall, the MLE estimates and the hBayesDM posterior means were **similar** across all three parameters. Most data points were located near the diagonal reference line, $y = x$,

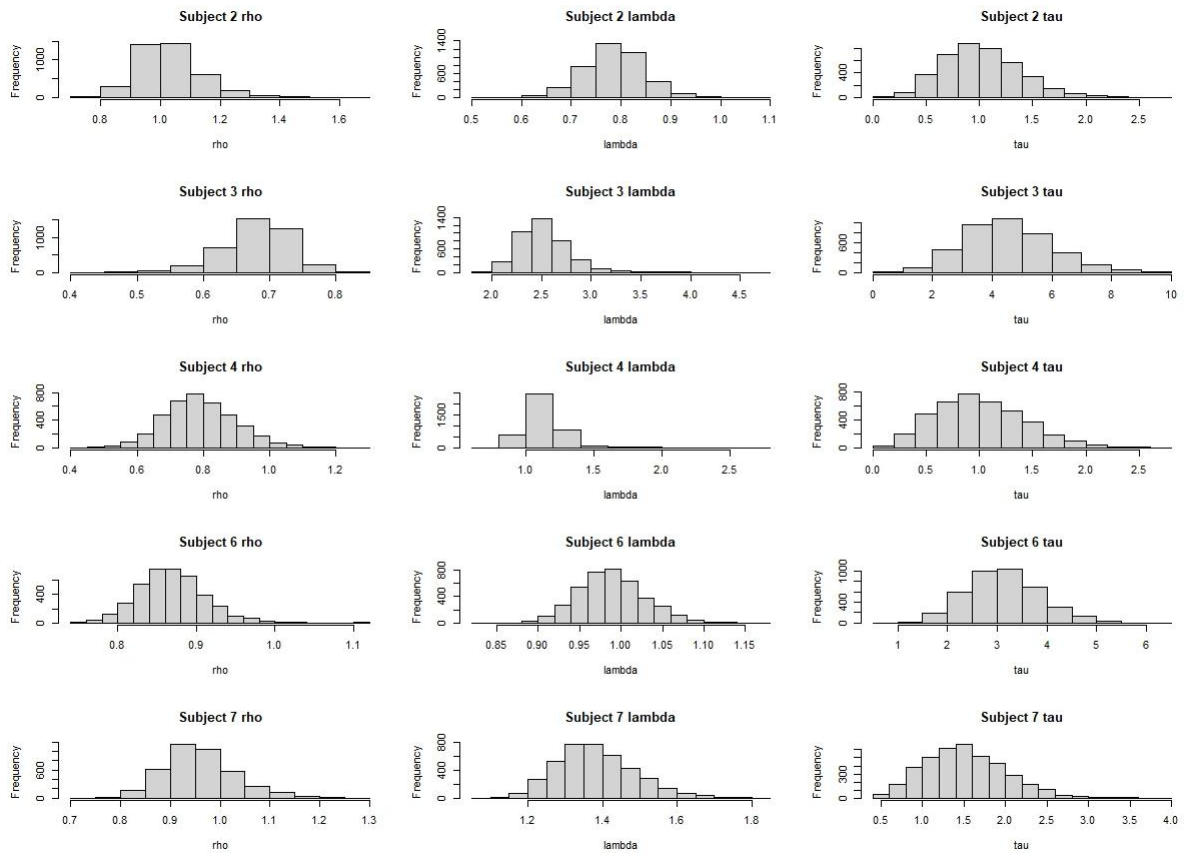
indicating strong agreement between the MLE estimates and the hBayesDM posterior means. However, posterior means tended to be **slightly shrunk** toward the group mean compared to the MLE estimates, especially for larger tau values.

1. (3) The MLE estimates and the hBayesDM posterior means differed slightly due to **shrinkage effects**. MLE estimates parameters independently for each subject and only maximizes the likelihood of the observed data. In contrast, hBayesDM uses hierarchical Bayesian estimation, which assumes that individual parameters are drawn from group-level distributions. As a result, parameter estimates are partially pooled toward the group mean, reducing extreme estimates and producing more stable estimates. In addition, Bayesian estimation incorporates **prior distributions** in parameter estimation, which can also contribute to differences from the MLE estimates.

2. (a)~(e)

2. (a) $BF_{12} \approx e^{(BIC_2 - BIC_1)/2}$	(e) $BF_{pow1i} \approx e^{(BIC_i - BIC_{pow1})/2}$
$\approx e^{(651.8656 - 580.4341)/2}$	$= e^{(BIC_1 - 475.292)/2}$
$= 3.2445 \times 10^{15}$	$BF_{pow1pow2} \approx e^{(477.371 - 475.292)/2} = 2.828$
(b) There is extreme evidence for model 1	$BF_{pow1exp1} \approx e^{(524.532 - 475.292)/2} = 4.924 \times 10^{10}$
(c) $\frac{P(M_1 D)}{P(M_2 D)} = \frac{P(M_1)}{P(M_2)} \times BF_{12}$	$BF_{pow1exp2} \approx e^{(489.535 - 475.292)/2} = 1238$
$= \frac{0.6}{0.4} \times 3.2445 \times 10^{15}$	$BF_{pow1expow} \approx e^{(479.451 - 475.292)/2} = 8.000$
$= 4.8668 \times 10^{15}$	$BF_{pow1hyp1} \approx e^{(482.954 - 475.292)/2} = 46.1086$
(d) $P(M_1 D) + P(M_2 D) = 1$	$BF_{pow1hyp2} \approx e^{(481.634 - 475.292)/2} = 23.8313$
$\frac{P(M_1 D)}{P(M_2 D)} = \frac{P(M_1 D)}{1 - P(M_1 D)} = 4.8668 \times 10^{15}$	
$P(M_1 D) = (1 - P(M_1 D)) \times 4.8668 \times 10^{15}$	
$P(M_1 D) = \frac{4.8668 \times 10^{15}}{1 + 4.8668 \times 10^{15}}$	
≈ 1	

3. (1)



3. (2)

```
> print(posterior_means)
  subjID      rho      lambda      tau
1      2 1.0296675 0.7855805 1.016122
2      3 0.6805573 2.5245039 4.554641
3      4 0.7902557 1.1306952 1.001212
4      6 0.8686806 0.9887838 3.126778
5      7 0.9588018 1.3830693 1.520071
```